

Module -2

Power BI Assignment 2 – DAX & Data Visualization

➤ TASK:

DAX Calculations

1) Calculated Columns:

- **Create a Calculated Column for 'Category Type':** Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into single 'Category Type' column.

DAX formula:

Category Type =
'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, displaying the formula bar with the DAX formula: `Category type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`. The 'Data' pane on the right shows the 'Order Details' table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and the newly created 'Column' (Category type). The table contains 20 rows of data.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Column
B-25602	561	212	3	Clothing	Saree	
B-25602	119	-5	8	Clothing	Saree	
B-25603	193	-166	3	Clothing	Saree	
B-25604	157	5	9	Clothing	Saree	
B-25605	75	0	7	Clothing	Saree	
B-25609	25	-5	4	Clothing	Saree	
B-25610	43	0	3	Clothing	Saree	
B-25611	160	-59	2	Clothing	Saree	
B-25613	1603	0	9	Clothing	Saree	
B-25619	353	90	8	Clothing	Saree	
B-25622	534	0	3	Clothing	Saree	
B-25623	149	-87	4	Clothing	Saree	
B-25625	635	-349	5	Clothing	Saree	
B-25628	24	-9	4	Clothing	Saree	
B-25633	711	-8	4	Clothing	Saree	
B-25635	382	30	3	Clothing	Saree	
B-25636	637	113	5	Clothing	Saree	
B-25640	122	-47	4	Clothing	Saree	
B-25646	20	-8	2	Clothing	Saree	
B-25647	42	-6	4	Clothing	Saree	
B-25648	55	-26	4	Clothing	Saree	

OUTPUT:

The screenshot shows the Power BI Desktop interface with the 'Category type' calculated column added to the 'Order Details' table. The formula bar shows the same DAX formula: `Category type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`. The 'Data' pane on the right shows the 'Order Details' table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and the newly created 'Category type'. The table contains 20 rows of data, with the 'Category type' column now populated with the concatenated values of 'Category' and 'Sub-Category'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type
B-25602	561	212	3	Clothing	Saree	Clothing_Saree
B-25602	119	-5	8	Clothing	Saree	Clothing_Saree
B-25603	193	-166	3	Clothing	Saree	Clothing_Saree
B-25604	157	5	9	Clothing	Saree	Clothing_Saree
B-25605	75	0	7	Clothing	Saree	Clothing_Saree
B-25609	25	-5	4	Clothing	Saree	Clothing_Saree
B-25610	43	0	3	Clothing	Saree	Clothing_Saree
B-25611	160	-59	2	Clothing	Saree	Clothing_Saree
B-25613	1603	0	9	Clothing	Saree	Clothing_Saree
B-25619	353	90	8	Clothing	Saree	Clothing_Saree
B-25622	534	0	3	Clothing	Saree	Clothing_Saree
B-25623	149	-87	4	Clothing	Saree	Clothing_Saree
B-25625	635	-349	5	Clothing	Saree	Clothing_Saree
B-25628	24	-9	4	Clothing	Saree	Clothing_Saree
B-25633	711	-8	4	Clothing	Saree	Clothing_Saree
B-25635	382	30	3	Clothing	Saree	Clothing_Saree
B-25636	637	113	5	Clothing	Saree	Clothing_Saree
B-25640	122	-47	4	Clothing	Saree	Clothing_Saree
B-25646	20	-8	2	Clothing	Saree	Clothing_Saree
B-25647	42	-6	4	Clothing	Saree	Clothing_Saree
B-25648	55	-26	4	Clothing	Saree	Clothing_Saree

➤ **Calculate Revenue per Order in Order Details Table:** Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

DAX formula:

Revenue per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]

The screenshot shows the Power BI Desktop interface. The formula bar at the top contains the DAX formula: `Revenue per order = 'Order Details'[Amount] * 'Order Details'[Quantity]`. Below the formula bar, the 'Columns' pane shows the calculated column 'Revenue per order' being added to the 'Order Details' table. The main view displays a table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Revenue per order. The table contains 1,500 rows of data.

OUTPUT:

The screenshot shows the final output table in Power BI Desktop. The table has columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Revenue per order. The table contains 1,500 rows of data. The 'Revenue per order' column is highlighted in green, indicating it is the selected column.

➤ **Create a Calculated Column to Categorize Sales:** Add a calculated column named

‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

DAX formula:

Sales Category =

IF ('Order Details'[Amount] > AVERAGE('Order Details'[Amount]), "Above Average",
"Below Average")

OUTPUT:

The screenshot shows the Power BI Desktop interface. The main view is a table visualization of the 'Order Details' table. A new column, 'sales category', has been added, calculated using the formula: `IF ('Order Details'[Amount] > AVERAGE('Order Details'[Amount]), "Above Average", "Below Average")`. The table lists various orders with columns for Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category type, Revenue per order, and the calculated sales category. The 'Sales category' column shows values like 'Above Average' and 'Below Average'.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue per order	sales category
B-25602	561	212	3	Clothing	Saree	Clothing_Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing_Saree	952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing_Saree	579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing_Saree	1413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing_Saree	525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing_Saree	100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing_Saree	129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing_Saree	320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing_Saree	14427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing_Saree	2824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing_Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing_Saree	596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing_Saree	3175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing_Saree	96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing_Saree	2844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing_Saree	1146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing_Saree	3185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing_Saree	488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing_Saree	40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing_Saree	168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing_Saree	220	Below Average

2) Calculated Measures

➤ **Order Count- Create a measure that calculates the total number of orders in the Order Details table.**

DAX formula:

{ Order Count = COUNT('Order Details'[Order ID]) }

OUTPUT:

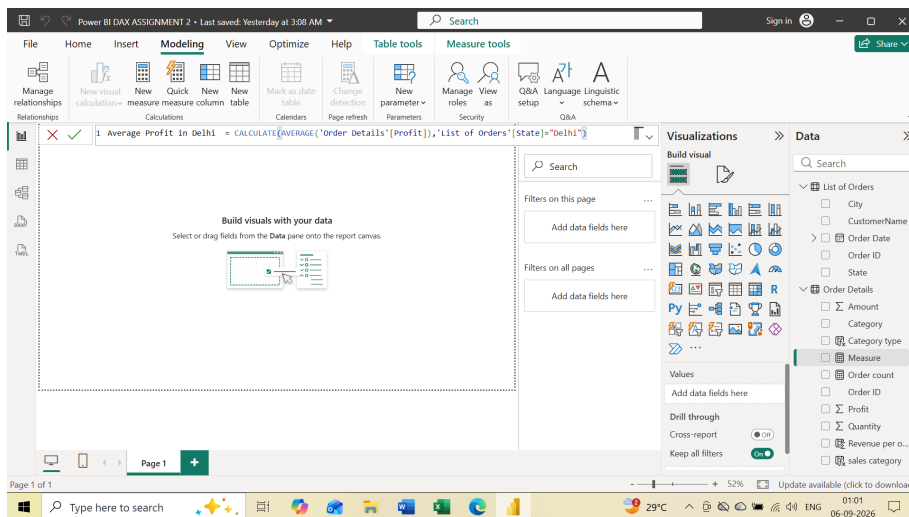
The screenshot shows the Power BI Desktop interface. The 'Measure' tab is selected, and the formula bar contains the DAX formula: `1. Measure = COUNT('Order Details'[Order ID])`. The 'Visualizations' pane on the right shows the 'Build visual' section with various chart types. The 'Data' pane on the right shows the 'Order Details' table with columns like City, CustomerName, Order Date, Order ID, State, Amount, Category, Category type, and Measure.

➤ **Average Profit in Delhi- Create a measure that calculates the average profit for orders placed in Delhi.**

DAX formula:

{ Average Profit in Delhi =CALCULATE(AVERAGE('Order Details'[Profit]),
'Order Details'[State] = "Delhi") }

OUTPUT:

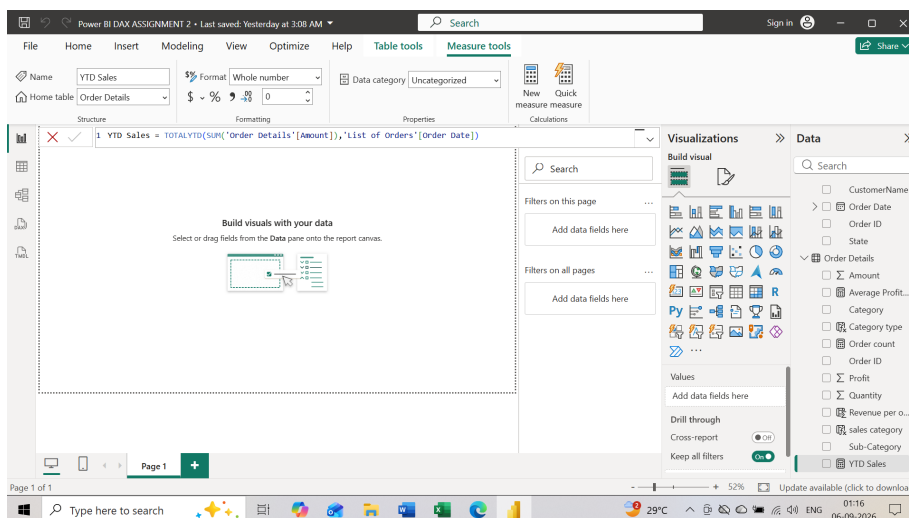


- **Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.**

DAX formula:

YTD Sales =TOTALYTD(SUM('Order Details'[Amount]),'List of Order'[Order Date])

OUTPUT:



3)Data Visualization

Create the following visualizations using Power BI:

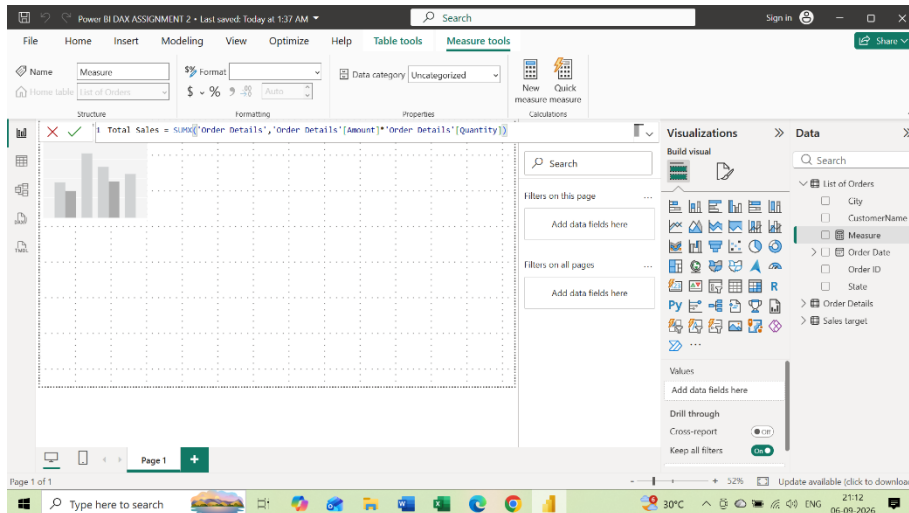
- **Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.**

Create Sales Target Achievement by Category :using a Clustered Column Chart.

i)Create the Actual Sales measure

DAX formula:

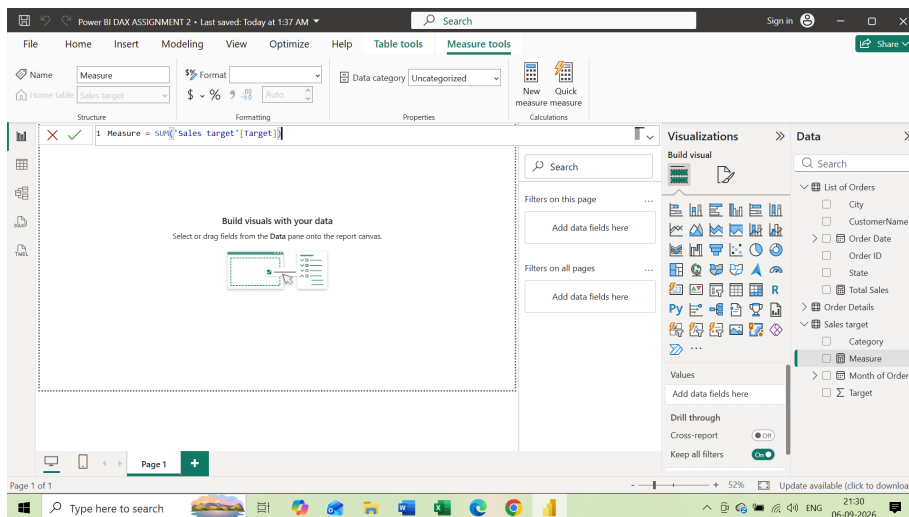
{ Total Sales = SUMX('Order Details','Order Details'[Amount] * 'Order Details'[Quantity]). }



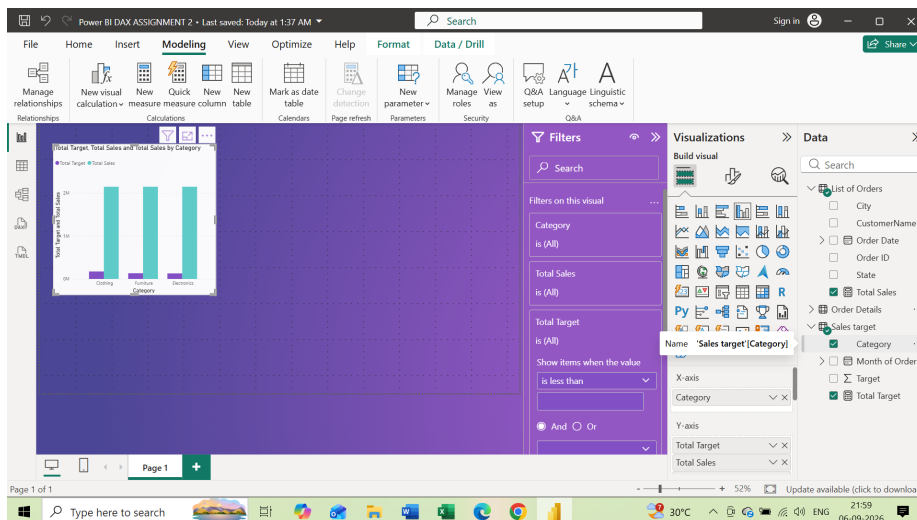
ii) Create the Sales Target measure

DAX formula:

{ Total Target = SUM('Sales Target'[Target]) }



OUTPUT:



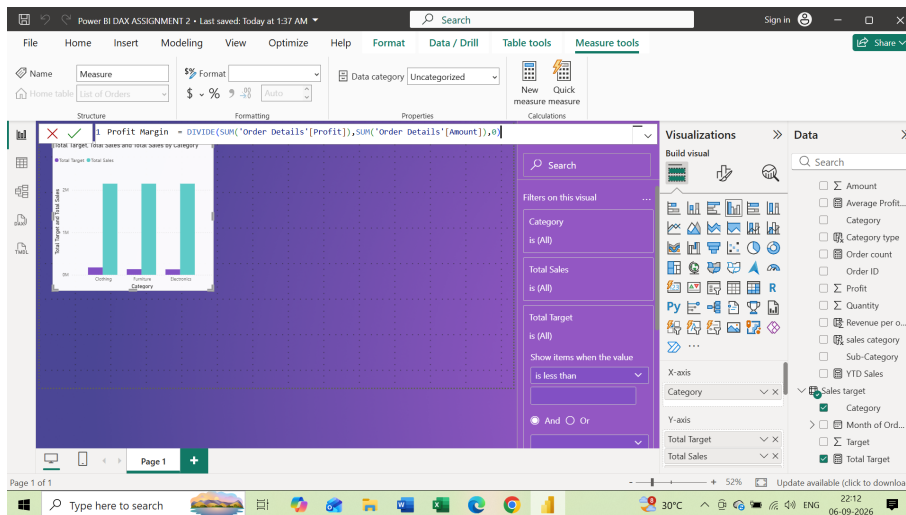
● **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

Create **Max Profit Margin by Sub-Category** in Power BI using a **Donut Chart**,

i) Create the Profit Margin measure

DAX Formula:

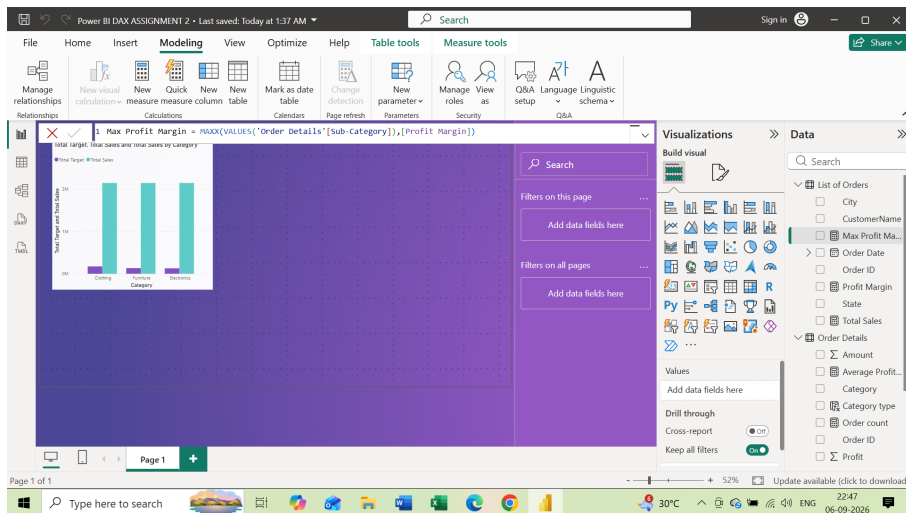
{Profit Margin = DIVIDE(SUM('Order Details'[Profit]),SUM('Order Details'[Amount]),0)}



ii) Create the Maximum Profit Margin measure

Create another measure:

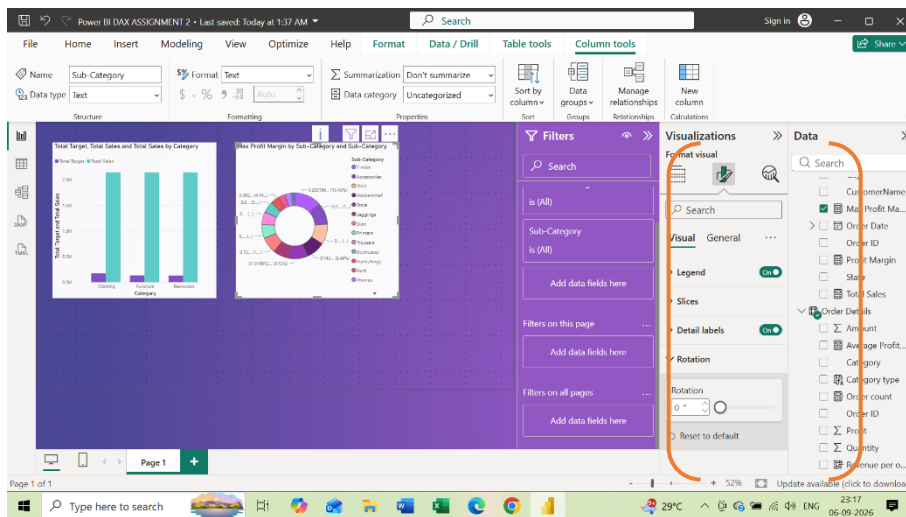
{Max Profit Margin = MAXX(VALUES('Order Details'[Sub-Category]),[Profit Margin])}



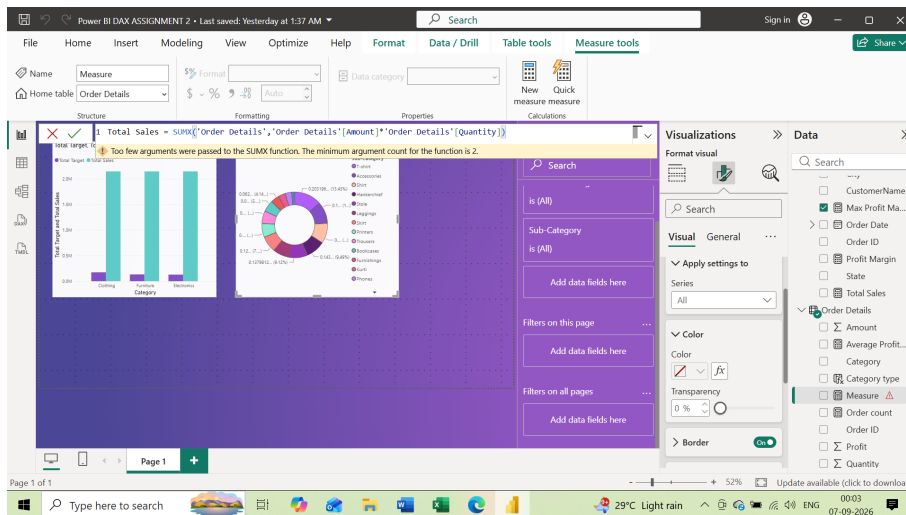
This calculates the **maximum profit margin for each Sub-Category**

iii) Create the Donut Chart and Format the chart

- **Sub-Category** → **Legend**
- **Max Profit Margin** → **Values**



OUTPUT:



Donut chart will show each **Sub-Category** as a slice, with the corresponding **maximum profit margin** displayed as a percentage.

- **Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.**

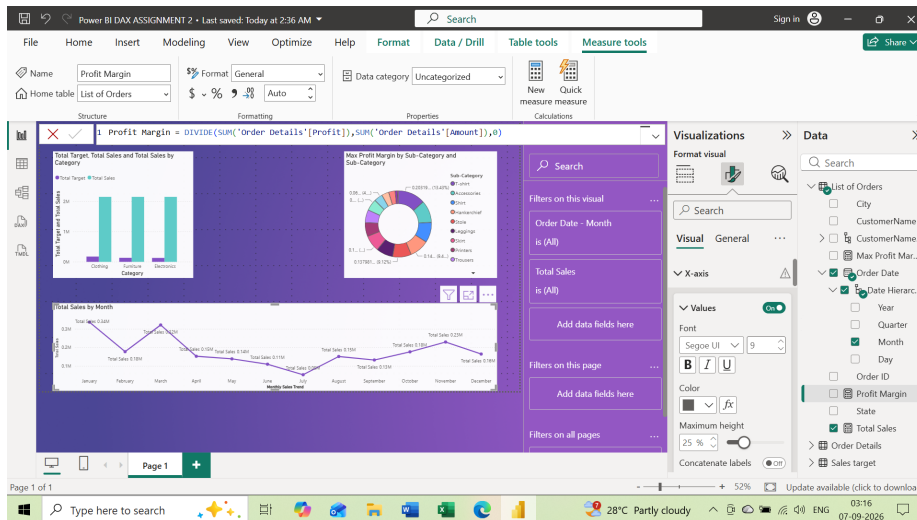
To create a **Monthly Sales Trend** in Power BI using a **Line Chart**, follow these steps:

i) Create a **Total Sales** measure

{Total Sales =SUMX('Order Details', 'Order Details'[Amount] * 'Order Details'[Quantity])}

ii) Insert the **Line Chart**, Set the date to **Month** & **Format the chart**

OUTPUT



● **Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.**

To create a **Comparison of Profit and Quantity by Sub-Category** using a **Scatter Chart**

i) **Create the Profit measure**

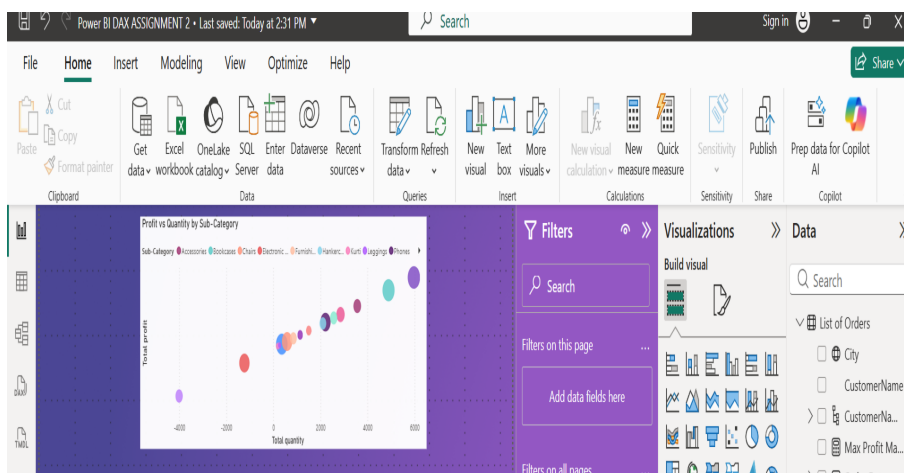
$$\text{Total Profit} = \text{SUM}('Order Details'[Profit])$$

ii) **Create the Quantity measure**

$$\text{Total Quantity} = \text{SUM}('Order Details'[Quantity])$$

iii) **Create the Scatter Chart, Format the chart then Format visual:**

OUTPUT:



Each **dot** represents a **sub-category**:

- **Right side** → higher quantity sold.
- **Higher position** → higher profit.
- **Top-right** → high quantity + high profit.
- **Bottom-right** → high quantity but relatively low/negative profit.
- **Top-left** → low quantity but high profit.

- **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

Need to create **two Card visuals** for Total Sales Amount and Sales Target, plus a **Multi-row Card** for the minimum target by segment.

i) Create Total Sales Amount measure:

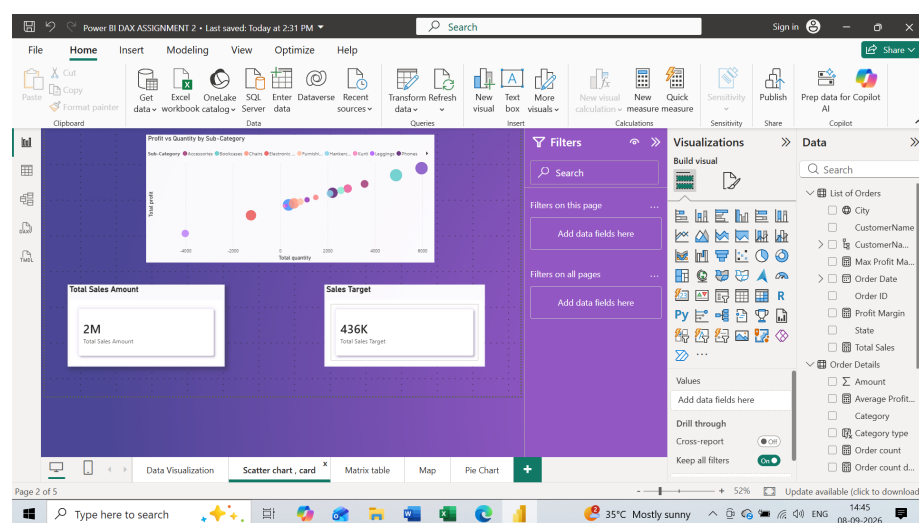
$$\text{Total Sales Amount} = \text{SUMX}(\text{'Order Details'}, \text{'Order Details'}[\text{Amount}] * \text{'Order Details'}[\text{Quantity}])$$

ii) Create Total Sales Target measure

$$\text{Total Sales Target} = \text{SUM}(\text{'Sales Target'}[\text{Target}])$$

iii). Create the Total Sales Amount Card and Sales Target Card:

OUTPUT:



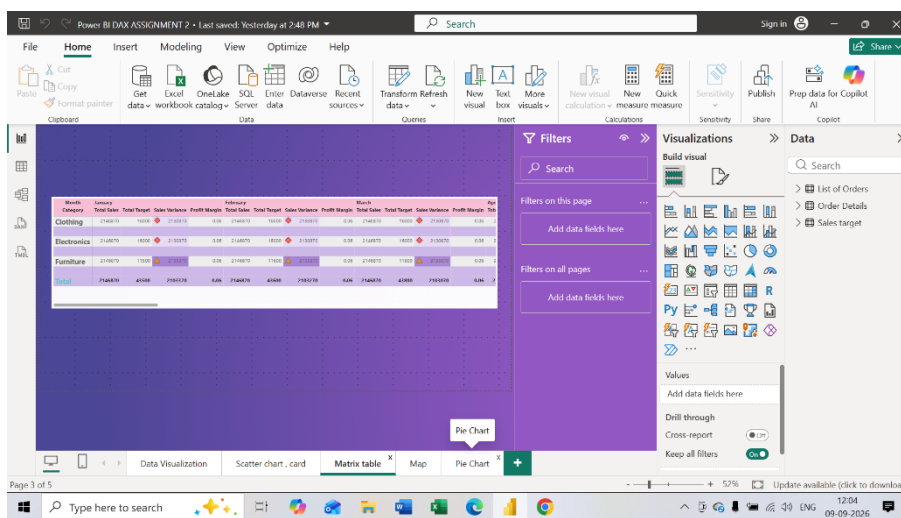
- **Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.**

To build a **Sales Performance Matrix** in Power BI that compares **Actual Sales vs Sales Target** by **Category and Month**

i) Create the Actual Sales measure

$$\text{Total Sales} = \text{SUMX}('Order Details', 'Order Details'[Amount] * 'Order Details'[Quantity])$$

OUTPUT:



- **Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.**

Create a **Geographic Sales Analysis** in Power BI and visualize **Total Sales by City** on a map,

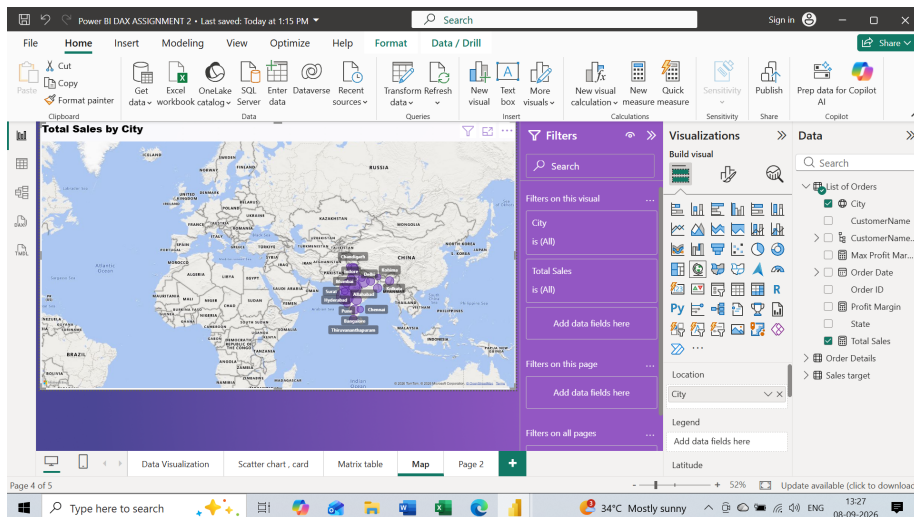
i) Already Created the Total Sales measure

ii) Insert a Map visual Put or Drag Total Sales → Bubble size.

iii) Set the City data category and Format the map

Title: **Geographic Sales Analysis – Total Sales by City**

OUTPUT:



Interpretation:

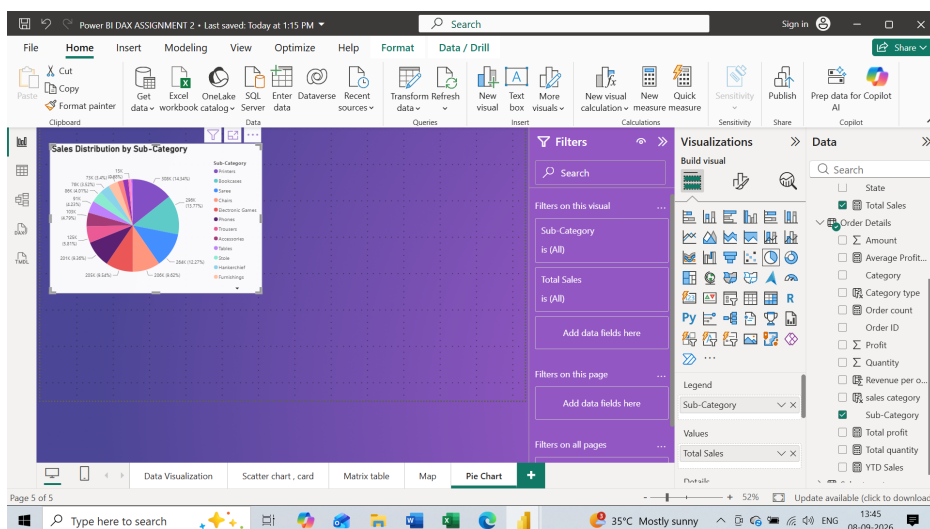
- **Larger bubbles** = higher total sales.
- **Smaller bubbles** = lower total sales.

• Sales Distribution by Sub-Category: Represent the sales distribution across Different.

Use the Total Sales measure and Create the visual Donut chart from the Visualizations pane.

i)The chart shows quick view of which **sub-categories** contribute the most and least to overall sales.

OUTPUT:



- **sub-categories using a tree map.**

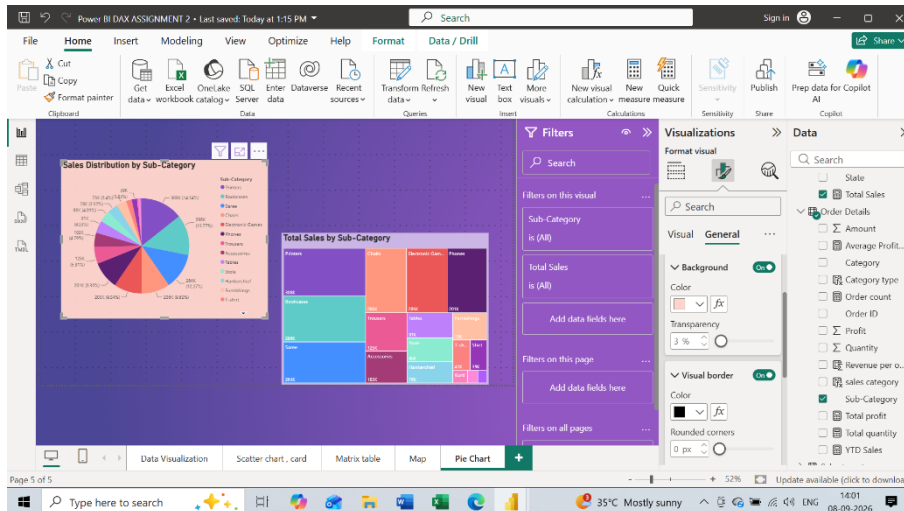
For Sales Distribution by Sub-Category using a Tree Map in Power BI:

i) Use the Total Sales measure

ii) Then Insert a Tree Map and Format the visual

The title: Sales Distribution by Sub-Category

OUTPUT:



Interpretation

The Tree Map displays each Sub-Category as a rectangle:

- **Larger rectangle** → higher sales
- **Smaller rectangle** → lower sales

This makes it easy to identify the **highest- and lowest-selling sub-categories** at a glance.

- **Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.**

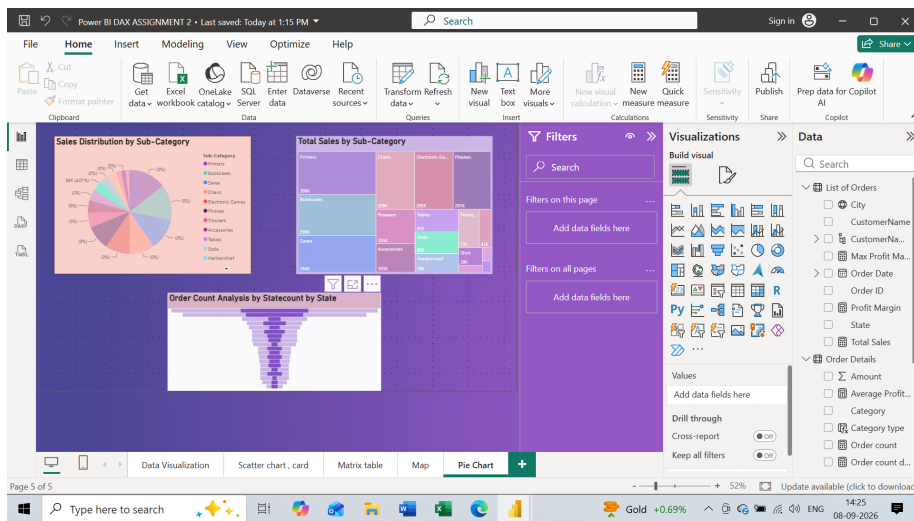
Create Order Count Analysis by State using a Funnel Chart in Power BI, follow these steps:

i) Use the Order Count measure

{Order Count =DISTINCTCOUNT('Orders Data'[Order ID])}

ii) Then Insert the Funnel Chart

OUTPUT:



Thus, have attached all the reports for the cleaned data. Hence Data analysed has been completed.